

Rayan Butt

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Studied **Computer Science at the University of York** specialising in **AI, computer vision, and full-stack development**. Experienced in **Deployment, Testing and Databases** with projects in **Go web scraping and analysis**, and **web application development**. Strong **problem-solving skills**, **adaptable in fast-paced environments**, and **eager to contribute to innovative projects and learn new skills**. Always ready to **take charge** and tackle problems with **unique and tailored approaches**.

Technical Skills

- **Languages:** SQL, Python, Go, TypeScript, JavaScript, Java
- **Frameworks & Libraries:** Pytorch, Flask, Node.js, React
- **Tools & Platforms:** Excel, Postgres, GitHub Actions (CI/CD), Linux, Windows, AWS.
- **Specialist Knowledge:** Front-end and back-end development, AI & Machine Learning, Data Structures & Algorithms, User-Centred Design, Network Administration.

Relevant Projects

Trading Card Price Tracker (Golang API) - <https://github.com/RayRB21/tcg-price-tracker>

Built a **Go CLI tool** that **reverse engineered PriceCharting's internal paginated JSON endpoint**, discovered via Chrome DevTools network analysis, to **scrape live prices across 700+ Riftbound trading cards**, persisting time-series snapshots to a **SQLite database** using versioned migrations. Implemented a pure **Go analysis engine** to **compute price movement** across stored snapshots, surfacing spike detection with a configurable threshold, top movers ranked by absolute percentage change, and per-card price history with moving averages. Designed a **cobra-powered CLI** with subcommands (scrape, movers, spikes, history) **enabling database queries** independent of the scraper, with idiomatic Go package structure across internal/scraper, internal/storage, and internal/analysis packages.

Sour Casa Bakery E-Commerce Website (React /Typescript /Flask) - <https://github.com/RayRB21/Portfolio-Website>

Built and deployed a full-stack e-commerce website for a family bakery using **React, TypeScript, and Flask**. Implemented a responsive user interface, dynamic product catalogue, shopping basket, Stripe payment integration, and delivery scheduling logic. Currently expanding the backend with a **PostgreSQL** database to replace hardcoded product data, enabling persistent product, order, and customer management through a REST API.

Budgeting Flask Web App (Flask/Postgres) - <https://github.com/RayRB21/SourCasa-Website>

Developed, deployed and maintained a personal budgeting web application which uses a linked **database**, enabling users to record income and expenses, set savings goals, and view spending trends. The project incorporates **Python, HTML/CSS, and Postgres for backend, frontend and data storage**, demonstrating skills in **full-stack development, API design, and database integration**. It also incorporates a **CI/CD Pipeline** with **GitHub Actions** to **automate testing** and improve deployment reliability.

Education

2021 – 2024 University of York

Diploma of Higher Education (2:1)

Relevant Modules:

- Data 2: Data Analysis and Management (**DATA**)
- Human-Computer Interaction 2: User Experience (**USER EXPERIENCE**)
- Intelligent Systems 1: Introduction to Artificial Intelligence (**AI**)
- Computer Vision & Graphics (**AI**)
- Evolutionary & Adaptive Computing (**AI**)

2013 – 2020 Parmiters School Watford

Employment History

Jan 2025 - Current Business Support Assistant Juste Macarons Ltd

- Assisted with **day-to-day business operations and administrative tasks**.
- Negotiated **client contracts contributing to over £100,000 in annual sales**.
- Maintained spreadsheets and organised business records **using Excel and Google Sheets**.
- Helped **communicate data to potential distributors and clients**.
- **Researched and contacted new business opportunities** and process improvements.

Jan 2025 - Dec 2025 Network Technician VRXTRA

- Managed a **10-node Windows-based VR lab environment** over a **TCP/IP LAN**, ensuring stable connectivity and low-latency performance across all machines.
- Configured and **maintained static IP addressing and network connectivity** for VR workstations, **resolving IP conflicts** and downtime issues.
- **Diagnosed and resolved** hardware, network, and VR device issues, including connectivity failures, driver faults, and peripheral (headset/USB) instability.
- **Monitored network performance and device status using administrative tools**, ensuring consistent uptime for VR session environments.

Jun 2023 - Current Tutor Self-employed/ Parmiters School

- Provided coding instruction to children at Parmiters and offered personalized tutoring to improve computer science grades.
- Improved student performance by **delivering private tutoring sessions with an average grade equivalent to an A**.
- Developed individualized lesson plans and explanations based on unique learning styles.